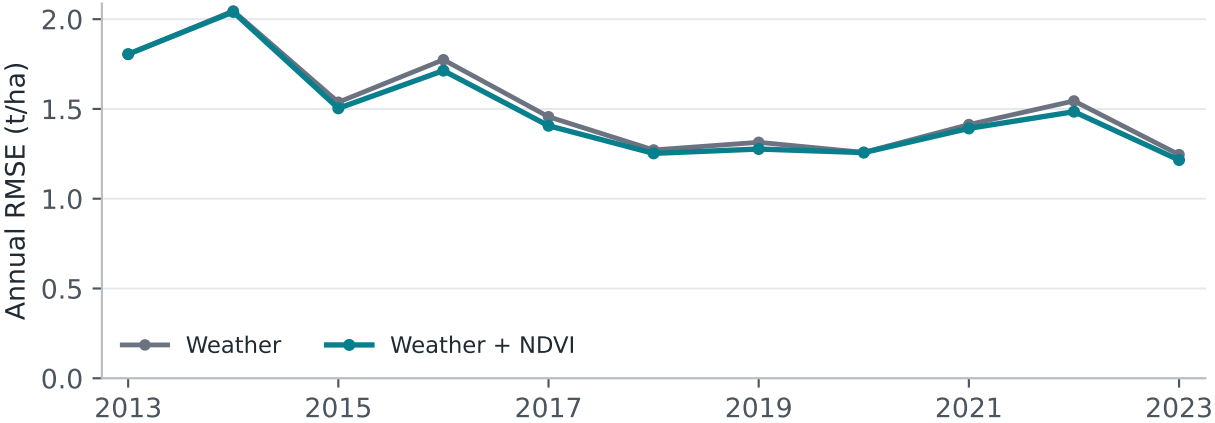


County maize yield: what vegetation adds to a weather forecast

US county forecasts, 2013-2023 · two fixed issue dates · current-vintage scientific test

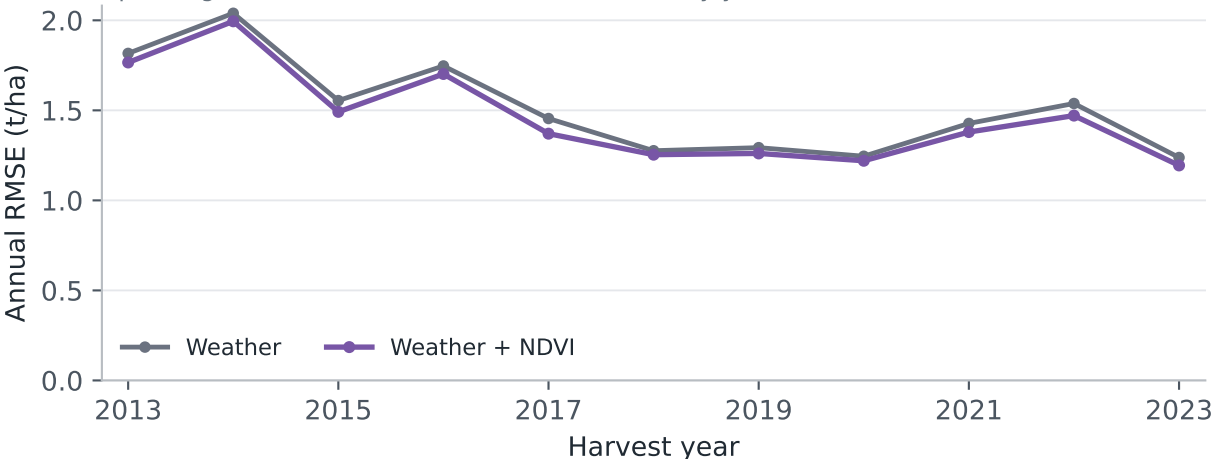
August 15 · primary

April-July observations · 15,026 scored county-years



September 15 · prespecified secondary

April-August observations · 14,907 scored county-years



Aug 15: +1.83% vs weather; frozen materiality gate FAIL | Sep 15: +3.12% vs weather; frozen materiality gate FAIL

Equal county weights within year and equal year weights. Horizons have different eligible support. Intervals use shared 5-calendar-year blocks (10,000 draws).

Revised archives and fixed 2021 crop geography; 2022/2023 are post-reference-year checks, not certified historical releases. Weather is AgERA5 reanalysis.

Exploratory tests; intervals are not adjusted across candidates. Forecast accuracy does not establish trading returns. Sources: USDA NASS / CY-Bench / NASA MODIS.

RMSE improvement over each comparator

Point estimates and paired 95% year-block intervals

